

AAPG2026	UP3DCYCLING	JCJCJ
Coordinated by :	Fabio A. CRUZ SANCHEZ	48 Months
Axe H.19 : Transformation des systèmes productifs de biens et de services - CES 10		

Distributed upcycling of mixed plastic wastes: material, process, and socio-technical challenges

UP3DCYCLING

I. Pre-proposal's context, positioning and objective(s)

Objectives of the project (O) and research hypotheses (H)

Mixed plastic waste remains a central issue for the circular economy of plastics. Although several studies argue that the mechanical recycling process is the most environmentally advantageous option¹, it has strong technical drawbacks when applied to multi-material/mixed plastic wastes. Currently, this type of plastics represent 30% of the plastic waste which is not managed by traditional recycling value chains, thus disposed for incineration. To address this, the focus of UP3DCYCLING is the development of methods and open source tools to allow the upcycling of mixed plastic wastes while reducing the environmental impact via local distributed recycling. Building upon the concept of mechanical upcycling understood as transforming disposable materials into objects of higher value²⁻⁴, ***the main goal is study, implement, and evaluate the upcycling potential of mixed plastic waste by in-situ reactive compatibilization via additive manufacturing.*** UP3DCYCLING conceptualizes this transformation across three perspectives. From a ***material perspective***, the first objective is to characterize the properties of mixed polymer waste blends to be usable AM feedstock **(O1)**. Blending approaches for post-consumer plastics upcycling have been studied incorporating compatibilizers for different mix of polymers to enhance the structural and mechanical properties of the resulting material **(H1)**. From a ***process perspective***, extrusion-based additive manufacturing—particularly open-source fused granular fabrication (FGF)—offers a promising pathway for multi-material processing **(H2)**. However, further adaptation of this technology is required to effectively handle mixed recycled polymers. Therefore, this project focuses on advancing and upgrading the FGF process to enable reliable and efficient processing of heterogeneous recycled materials **(O2)**. Finally, from a ***design for sustainability (DfS) perspective***⁵, the goal is to test a scalable and economically viable upcycling process, crucial for the widespread adoption of recycled materials in additive manufacturing **(O3)**. Distributed recycling approach involves the integration of the material and process with the product/system service by establishing small-scale facilities within local communities. By operating closer to the source of waste generation, these localized systems reduce the need for extensive transportation and promote greater community participation in the recycling process **(H3)**. Thus, UP3DCYCLING offers an integrated approach combining material, process and socio-technical levels to propose new recycling value chains in the service of a local circular economy.

Position of the project in relation to the state of the art

Global context

Plastic waste remains a global challenge, with no international agreement yet on reducing virgin plastic production. As a result, global output reached 413 Mt in 2023, with less than 10% derived from recycled materials—mostly single polymers such as PET. This shows that conventional recycling systems are inefficient from an economic and environmental perspective. A systemic approach capable of valorizing mixed and multi-material plastic waste is essential to significantly enhance material recovery rates.

On the other hand, Additive manufacturing (AM) technology makes possible the creation of complex geometries directly from digital designs. Indeed, Distributed Recycling via Additive Manufacturing (DRAM) is an emerging approach that redefines production and recycling systems^{6,7}. It offers new opportunities at the plastic value chains, bringing recycling to local contexts as demonstrated by fablabs and community spaces. By repurposing plastic waste into AM feedstock, it transforms discarded materials into valuable products, supporting a circular economy⁸. This model empowers communities to manage plastic waste more effectively while fostering local economic development through small-scale production and innovation. Our previous results showed that the DRAM approach reduces transportation costs and decreases environmental impacts (up to 97%) compared to centralized recycling systems^{9,10}. However, despite its promising potential, DRAM still faces significant challenges, including the degradation of polymer properties over multiple recycling cycles, difficulties in processing heterogeneous plastic mixtures, and issues with the reproducibility and scalability of results across different socio-technical contexts. The use of mixed plastic waste as feedstock for AM is a step forward towards a circular economy^{11,12}.

Specific context

While the implementation of recycled mixed plastic in AM is a promising and rapidly advancing field¹³, it presents major challenges: (1) material compatibility, (2) processing and technological limitations, and (3) global environmental assessment to connect with regulatory frameworks.

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First, the non compatibility between the various chemical and molecular structures from different plastics is a drawback due to their inherently lower interphase tension and lack of chemical compatibility, resulting in low-quality materials¹¹. In this regard, several strategies have been implemented such as ad-hoc process technologies¹⁴, the incorporation of compatibilizers¹⁵ such as, multi-block copolymers¹⁶ and nanofillers¹⁷. Our recent literature showed that chemical compatibilizers facilitate the blending of different polymers, opening new possibilities for reusing waste materials in AM applications¹⁸. Nevertheless, there remains a gap in the literature in terms of the validation of these new compatibilizers. Indeed, there are no standard methods to evaluate the quality of a blend composite¹⁹. A systematic benchmarking assessment to create minimal standards of quality in the AM process domain using compatibilizers for blended recycled material is still lacking in the scientific domain.

Likewise, fused granular fabrication (FGF) has emerged as an advanced AM technology that, unlike fused filament fabrication (FFF), utilizes polymer pellets directly fed into the system through a screw-based extruder²⁰. It is a promising technology for recycling waste, as it enables the direct use of recycled polymers reducing the conventional melt–solidification cycle, thereby improving material longevity and reducing costs^{7,13}. However, it is challenging, particularly in achieving consistent, reliable printability. For mixed plastics, achieving adequate homogenization of blends remains a significant challenge. The recent integration of twin-screw or planetary roller extruders into open-source additive manufacturing printers offers a technical pathway to improve blend uniformity and overall material performance²¹. To the best of the authors’ knowledge, this approach has not yet been fully developed.

Lastly, at the socio-technical level, comprehensive assessments of business models and social engagement are needed to better understand the transition from a techno-centric to a human-centric framing of DRAM. Although several life-cycle assessments (LCAs) have examined distributed recycling^{22,23}, the field remains at an early stage in addressing its major implications as a sustainability transition beyond technological performance²⁴. The development of adequate circular business models and the community participation to collect and manage waste at local scale are still underexplored. To advance this understanding, hybrid modeling approaches that integrate LCA, circular economy metrics²⁵, and multi-criteria methods should be developed to have a holistic evaluation. Despite its potential, DRAM faces challenges to widespread adoption. Thus, strengthening community-driven recycling ecosystems is essential to enable the systemic transition of local manufacturing toward circularity.

Methodology to reach the scientific objectives of the project

The UP3DCYCLING project brings together multidisciplinary expertise across three interconnected levels—material, process, and socio-technical—supported by experimental, modeling, and simulation approaches. Together, these dimensions form a comprehensive framework for evaluating and scaling distributed upcycling through open-source fused granular fabrication (FGF). The integration and interaction among these three levels represent a key added value of the project, enabling a holistic understanding of both technical and social dynamics. In this context, we propose to study:

WP1 – Material Characterization and Open Database Building

At the material level, the objective is to establish an open-source database integrating the technical requirements for blending common post-consumer waste. Special attention will be given to the role of compatibilization in improving interfacial adhesion between immiscible polymers. This work seeks to quantify in what extent compatibilization process can enhance blend homogeneity and printing quality, ultimately creating an open-source database of formulation and processing parameters for multi-material blends. In **task 1.1** a systematic review will be performed using PRISMA protocol to collect, classify, and analyze existing studies on multi-material polymer blends, building on previous works^{13,18}. The review will identify compatible material combinations, compatibilization strategies, and additive systems relevant for distributed recycling. In **task 1.2**, an experimental characterization plan will be conducted in laboratory-scale to evaluate blends with and without compatibilizers. The assessment will include chemical analyses (FTIR and Raman spectroscopy) to determine molecular interactions, thermal test (DSC, TGA), and rheological tests to evaluate flow and printability behavior. Then, **task 1.3** will analyze mechanical and morphological assessment through tensile, impact, and morphological analysis, using injection molding as a baseline for comparison with FGF-produced samples. All results will be structured in an open, multi-criteria database, following open-science principles to facilitate reproducibility and community use.

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WP2 – Open source Fused Granular Fabrication Development

At the process level (**WP2**), the goal is to design and validate an open-source FGF system (twin-screw / planetary roller) capable of handling heterogeneous waste blends. Furthermore, a specific focus will be made in the integration of real-time monitoring and closed-loop control of printing conditions to reduce defects such as warping or under-extrusion, following the product-driven control principles²⁶. Building on the open-hardware ecosystem, **task 2.1** will be concentrated on system design and development specifically to treat blend wastes. We will design and prototype an extruder-based FGF system, integrating mechanical and thermal control subsystems optimized for polymer blend processing. Then, **task 2.2** is focused on real-time process monitoring integration. We will implement in-situ defect detection and self-supervised digital twin models for adaptive process control following recent developments in digital manufacturing²⁷. These models will enable predictive adjustment of process parameters to mitigate defects such as under-extrusion, over-extrusion, warping, or layer delamination. Afterwards, **task 2.3** is set up to implement a benchmarking phase to compare FGF performance against standard extrusion and FFF techniques to identify process-specific limitations and opportunities. Finally, an in-situ product-driven control loop will be deployed to enable real-time adaptive printing, enhancing reliability and reproducibility of the FGF process. Finally, in **task 2.4** an open-source repository containing datasets, experimental protocols, and design files to ensure transparency, reproducibility, and accessibility for the wider research and maker communities.

WP3 – Design for sustainability: Circular business model and pluralistic evaluation

At the design for sustainability level, we will model the broader environmental, economic, and social implications of DRAM. **Task 3.1** will establish systems dynamics model to understand the structure of the distributed recycling approach. Then, **task 3.2**, will be dedicated to the analysis of circularity indicators and life-cycle assessment (LCA) methods to connect material recycling processes with industrial and local ecosystems. Finally, **task 3.3** will focus on structuring a circular business model identifying key technical, economical and social dimensions to support evidence-based strategies for circular economy implementation, improving the technical maturity of DRAM.

WP4 – Experimentation and deployment in local territory

Finally, **WP4** focuses on the iteration, validation, and territorial deployment of the developed materials and technologies. **Task 4.1** Real-world case studies will be conducted in collaboration with local stakeholders at Nancy around our experimental platform ‘Green Fablab’ to evaluate the performance, scalability, and social acceptance of distributed upcycling. **Task 4.2** will integrate the outcomes to generate feedback loops to refine material formulations, process parameters, and socio-technical integration strategies, ensuring both replicability and community-driven adoption. Moreover, the results of this evaluation will provide valuable insights into the key drivers and barriers influencing the implementation and scalability of local recycling ecosystems, shedding light on the technical readiness level.

Added-value in terms of scientific contribution

The UP3DCYCLING ground-breaking scientific nature relies on how multi-material plastic waste can be effectively processed through open-source additive manufacturing (AM) to enable a sustainable distributed upcycling approach. Connecting the embedded levels of material compatibilization, AM process optimization and socio-technical dimensions, UP3DCYCLING seeks that AM play a key role for circular economy by using locally sourced plastic waste.

Ability of the project to address the research issues covered by the chosen research theme

UP3CYCLING involves creating integrated workflows that can efficiently blend and process recycled plastic wastes for a production system for circular economy. Bridging the gap between material development, process innovation, and socio-technical evaluation will offer a comprehensive understanding of how distributed upcycling via open-source FGF can support a viable, sustainable, and community-centered circular economy. This is a major step forward in the localized, circular, and innovative manufacturing paradigm of the future.

II. Partnership

Scientific coordinator. Fabio A. Cruz Sanchez (<https://orcid.org/0000-0001-8529-5327>) (Phd Defence: 12/2016, Recruitment 09/2023) is assistant professor in mechanical engineering at the University Lorraine / ERPI (EA 3767). He will engage 67% of his research time (32 PM). Since 2017, he has been implementing an urban living lab distributed recycling demonstrator called ‘Green Fablab’ (<https://lf2l.fr/projects/green-fablab>) at Nancy, France as a research-action perspective. He has coordinated tasks in H2020 Project INEDIT (GA: 869952) and ANR GreenLocal3D. Also, he has managed EU Erasmus+ projects Climatelabs, and currently is coordinating EU Erasmus+ KA2 TechTraplastiCE. The current proposition offers an excellent opportunity to build up a solid research network for an

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ERC project in the future to better understand distributed recycling as a complex socio-technical system for circular economy in europe - latin-american context.

Scientific coordinator's team: WP1 will include **Prof. Cécile Nouvel** (Université de lorraine, LRGP -UMR 7274-) specialist in polymer chemistry and formulation and **Prof. Sandrine Hope** CR CNRS (LRGP), specialist in compounding and processing technologies. In WP2, **Prof. Joshua Pearce** (Western University - Canada) will bring his expertise in open-source hardware and **Prof Hakim Boudaoud** (UL - ERPI) will help in its experiences with distributed recycling. Then in WP3, **Prof. Andrea Espinoza** (USACH - Chile) will add her experience on multi-objective optimization, life cycle analysis, supply chain design, and sustainability. Additionally, **Baptiste Monsaingeon** (CNRS – LISIS & Université de Reims Champagne-Ardenne) will bring his expertise as a sociologist to understand the social representations and practices of the community to foster small-scale production and innovation ecosystems. For WP4, **Prof. Mauricio Camargo** (UL - Director ERPI) will add his expertise on multi-criteria evaluation and technology management and product-process design/integration. This project will involve 1 PhD student (1 in WP1/WP2) and 24-month postdoc (WP3/WP4) and 4 M2 trainees.

III. References

- Klotz, M., Haupt, M. & Hellweg, S. **Potentials and limits of mechanical plastic recycling.** *Journal of Industrial Ecology* **n/a**, (2023).
- Muzata, T. S., Matuana, L. M. & Rabnawaz, M. **Challenges in the mechanical recycling and upcycling of mixed postconsumer recovered plastics (PCR): A review.** *Current Research in Green and Sustainable Chemistry* **8**, 100407 (2024).
- Hassan, M., Mohanty, A. K. & Misra, M. **3D printing in upcycling plastic and biomass waste to sustainable polymer blends and composites: A review.** *Materials & Design* **237**, 112558 (2024).
- Jehanno, C. *et al.* **Critical advances and future opportunities in upcycling commodity polymers.** *Nature* **603**, 803–814 (2022).
- Ceschin, F. & Gaziulusoy, İ. *Design for Sustainability: A Multi-level Framework from Products to Socio-technical Systems.* (Routledge, London, 2019). doi:10.4324/9780429456510.
- Little, H. A. *et al.* **Towards Distributed Recycling with Additive Manufacturing of PET Flake Feedstocks.** *Materials* **13**, 4273 (2020).
- Romani, A., Levi, M. & Pearce, J. M. **Recycled Polycarbonate and Polycarbonate/Acrylonitrile Butadiene Styrene Feedstocks for Circular Economy Product Applications with Fused Granular Fabrication-Based Additive Manufacturing.** (2023) doi:10.2139/ssrn.4508039.
- Cruz Sanchez, F. A., Boudaoud, H., Camargo, M. & Pearce, J. M. **Plastic recycling in additive manufacturing: A systematic literature review and opportunities for the circular economy.** *Journal of Cleaner Production* **264**, 121602 (2020).
- Santander, P., Cruz Sanchez, F. A., Boudaoud, H. & Camargo, M. **Closed loop supply chain network for local and distributed plastic recycling for 3D printing: A MILP-based optimization approach.** *Resources, Conservation and Recycling* **154**, 104531 (2020).
- Caceres-Mendoza, C. *et al.* **Life cycle assessment of filament production in distributed plastic recycling via additive manufacturing.** *Cleaner Waste Systems* **5**, 100100 (2023).
- Kassab, A., Al Nabhani, D., Mohanty, P., Pannier, C. & Ayoub, G. Y. **Advancing Plastic Recycling: Challenges and Opportunities in the Integration of 3D Printing and Distributed Recycling for a Circular Economy.** *Polymers* **15**, 3881 (2023).
- Byard, D. J. *et al.* **Green fab lab applications of large-area waste polymer-based additive manufacturing.** *Additive Manufacturing* **27**, 515–525 (2019).
- Gonzalez, C. S. *et al.* **Strategies for recycling multi-material polymer blends for additive manufacturing.** *Sustainable Materials and Technologies* **44**, e01430 (2025).
- Potiyaraj, P., Tanpichai, S. & Phanwiroj, P. **Physical Properties of PP/Recycled PET Blends Prepared by Pulverization Technique.** *Advanced Materials Research* **488–489**, 109–113 (2012).
- Ghosh, A. **Performance modifying techniques for recycled thermoplastics.** *Resources, Conservation and Recycling* **175**, 105887 (2021).
- Nomura, K. *et al.* **Multiblock Copolymers for Recycling Polyethylene–Poly(ethylene terephthalate) Mixed Waste.** *ACS Applied Materials & Interfaces* **12**, 9726–9735 (2020).
- Liu, Y. *et al.* **Effect of nanoparticles on the morphology and properties of PET/PP in situ microfibrillar reinforced composites.** *Polymer Composites* **38**, 2718–2726 (2017).
- Suescun Gonzalez, C., Cruz Sanchez, F. A., Boudaoud, H., Nouvel, C. & Pearce, J. M. **Multi-material distributed recycling via material extrusion: Recycled high density polyethylene and poly (ethylene terephthalate) mixture.** *Polymer Engineering & Science* **n/a**, (2024).
- Zander, N. E. **Recycled Polymer Feedstocks for Material Extrusion Additive Manufacturing.** in *ACS Symposium Series* vol. 1315 37–51 (American Chemical Society, 2019).
- Alexandre, A., Cruz Sanchez, F. A., Boudaoud, H., Camargo, M. & Pearce, J. M. **Mechanical Properties of Direct Waste Printing of Polylactic Acid with Universal Pellets Extruder: Comparison to Fused Filament Fabrication on Open-Source Desktop Three-Dimensional Printers.** *3D Printing and Additive Manufacturing* 3dp.2019.0195 (2020) doi:10.1089/3dp.2019.0195.
- Formela, K. & Eyigöz, B. **Planetary roller extruders in the sustainable development of polymer blends and composites – Past, present and future.** *Express Polym. Lett.* **18**, 441–458 (2024).
- Kerdlap, P. *et al.* **Environmental evaluation of distributed versus centralized plastic waste recycling: Integrating life cycle assessment and agent-based modeling.** *Procedia CIRP* **90**, 689–694 (2020).
- Kerdlap, P. *et al.* **Comparing the environmental performance of distributed versus centralized plastic recycling systems: Applying hybrid simulation modeling to life cycle assessment.** *Journal of Industrial Ecology* **26**, 252–271 (2022).
- Eising-Mertsch, L., Savaget, P., Zomer, T. & Brophy, A. **Frames of systems change in sustainability transformations: Lessons from sociotechnical systems and circular economy case studies.** *Journal of Cleaner Production* **479**, 143976 (2024).
- Saidani, M. *et al.* **The future of circular economy metrics: Expert visions.** *Resources, Conservation and Recycling* **205**, 107565 (2024).
- Pronost, G., Mayer, F., Camargo, M. & Dupont, L. **Digital Twins along the product lifecycle: A systematic literature review of applications in manufacturing.** *Digital Twin* **1**, 3 (2024).
- Deng, X. *et al.* **In-situ real-time defect detection, mitigation and self-supervised adaptation based on visual foundation model for material extrusion additive manufacturing.** *Additive Manufacturing* 104978 (2025) doi:10.1016/j.addma.2025.104978.